Temi Otun

otun@ualberta.ca | GitHub | LinkedIn | Website

Education

University of Alberta — Edmonton, AB

Bachelor of Science, Computing Science – Artificial Intelligence Specialization, Minor in Business — *Expected May 2027*

- **Awards:** Jason Lang Scholarship, Dean's List

Experience

Data Engineering Intern – Applied Machine Learning — May 2026 – Aug. 2026

Royal Bank of Canada (RBC) – Amplify Program — *Vancouver, BC*

- Co-developed a **patent-pending fraud detection system** combining ML classification with LLM analysis, achieving **94% recall** and an **85% F1-score**, projected to mitigate **$18M in annual fraud losses**
- Engineered a cascaded NLP pipeline processing **23M+ annual unstructured text records** using semantic embeddings and multi-stage classification, reducing operational costs by **65%** (**$1.5M in projected annual savings**)
- Led model experimentation across **6+ ML architectures**, optimizing decision thresholds and ensemble strategies that improved precision from **49% to 77%** with the resulting system projected to protect **12K+ clients annually**

Machine Learning Research Intern — Jan. 2025 – Aug. 2025

The Metabolomics Innovation Centre — *Edmonton, AB*

- Enhanced long-range temperature forecasting system, resulting in up to **47%** lower MAE with an average **24.3%** improvement across a six-year test set compared to previous approaches
- Developed multivariate time series models for precipitation forecasting, incorporating **100+** climate features and outperforming baseline models in **73%** of test years
- Assembled forecasting pipelines with advanced feature engineering and benchmarked **20+** ML models, attaining a **42%** accuracy improvement over baseline methods

Machine Learning Research Assistant — Sept. 2024 – Apr. 2026

University of Alberta — *Edmonton, AB*

- Developed ML systems for dementia and mild cognitive impairment detection using Whisper-derived speech embeddings and ensemble classifiers, contributing to research presented at ICASSP 2025
- Built survival-analysis and clinical prediction pipelines on large-scale MIMIC-IV patient records, applying feature engineering, temporal preprocessing, and rigorous model evaluation
- Designed and evaluated **15+** classification and regression experiments across speech, ECG, and clinical datasets using hyperparameter optimization and statistical performance analysis

Data Management Intern — Jan. 2024 – May 2024

InfoStrux — *Vancouver, BC*

- Optimized **25+** SQL queries in Snowflake, reducing execution time by up to **60%** and boosting performance of business intelligence dashboards
- Partnered with a senior data engineer to design Snowflake staging and curated layers for **3** datasets; wrote and tuned **30+** queries, improving data reusability and cutting time-to-insight by **40%**

Projects & Leadership

Emotion Detection CNN | *PyTorch, ML, Computer Vision, AI, Data Processing, Python* — GitHub

- Trained a deep convolutional neural network in PyTorch on the FER-2013 dataset with **32K+** labeled images for multi-class facial emotion recognition
- Achieved **70%** test accuracy across **7** emotion classes and integrated webcam inference for real-time emotion detection

Event Lottery System | *Java, Kotlin, Firebase, Gemini API, Google Maps* — GitHub

- Collaborated in a **seven-person Agile team** to build a Firestore-backed Android platform supporting **3 user roles** and end-to-end lottery workflows, including waitlists, randomized selection, invitations, and automatic redraws
- Integrated QR-code registration, geolocation, and real-time notifications with a serverless **Gemini-powered classification pipeline** that mapped user-generated event tags across **21 validated categories**

VP, Member Outreach & Community Engagement — Aug. 2026 – Present

ColorStack — *Edmonton, AB*

- Lead campus-wide outreach and recruitment initiatives, building partnerships with student organizations to expand membership

Technical Skills

Programming: C, Java, Python, SQL
ML/AI: Deep Learning, Reinforcement Learning, NLP, Computer Vision, Time Series, Generative AI, Statistical Learning, Representation Learning, Feature Engineering, Model Evaluation
Libraries & Frameworks: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, OpenCV, XGBoost, LightGBM, CatBoost, Hugging Face
Tools & Platforms: PostgreSQL, Snowflake, Firebase, Git, Docker, Kubernetes, Linux, AWS S3